

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

DECEMBER/JANUARY EXAMINATION

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS

TUESDAY, 30 DECEMBER 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 1**

CampusMart is a small online bookstore that sells around 200 books. The owner plans to expand the business and expects to handle thousands of book records. The system developer must choose the most efficient search algorithm to make sure users can find books quickly even when the database grows larger.

- State and explain **THREE (3)** characteristics of an algorithm for a large-scale system like CampusMart. (6 marks)
- Identify and explain the **TWO (2)** main types of complexity that the team should consider when selecting algorithms for online bookstore system. (8 marks)
- Relate the running time analysis with the algorithm selection. (2 marks)
- Table 1-1** shows the running time functions of three sorting growth rate — P, Q and R — as a function of input size n .

Algorithm	Growth-rate function, $f(n)$
P	$5n^2 + 20n + 100$
Q	$40n\log n + 600$
R	$150n + 90$

Table 1-1: Growth rate functions for algorithm P, Q and R

Determine the Big-O notation for algorithms P, Q, and R. Show your working clearly.

(9 marks)

[Total: 25 marks]

Question 2

BrightNote Stationery Shop has two sales counters, CounterA and CounterB. Each counter stores a range of stationery items that customers can purchase. To simplify inventory management, easier the shop uses a Set data structure to keep track of item names. Current items at each counter:

- CounterA: “notebook”, “pencil”, “highlighter”
- CounterB: “pencil”, “eraser”, “marker”

The shop manager wants to analyse items that are available in both counters and items that are unique to each counter, to plan promotions and restocking.

- Describe the **TWO (2)** advantages of using a Set instead of an Array to store the items at the counters. (4 marks)

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 2 (Continued)**

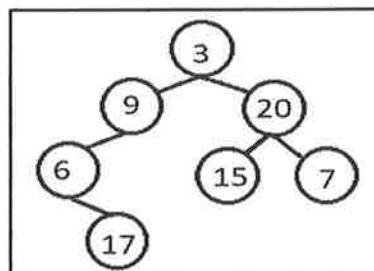
- b) In the BrightNote Stationery Shop system, compare HashSet and LinkedHashSet based on item ordering and how efficient they perform when checking or managing items. (4 marks)
- c) Write a Java program that performs the following tasks using HashSet:
- Create two HashSet objects named counterA and counterB.
 - Add the items listed above to each counter.
 - Find the items that are unique to CounterA (items in CounterA but not in CounterB).
 - Display the results using a loop:
 - Print all items unique to CounterA.
 - Print the total number of unique items in CounterA.

(17 marks)

[Total: 25 marks]

Question 3

- a) The library can implement the book records using either an array or a linked list. Explain the differences between arrays and linked lists in terms of memory allocation, ease of insertion/deletion, and access time. (6 marks)
- b) A student wants to implement a task scheduler. Explain whether a stack or a queue is more suitable and explain the reason. (6 marks)
- c) A social networking app models friendships between users, where each user can connect with multiple others. Determine whether a tree or a graph is the most suitable data structure to represent these connections, and justify your choice. (6 marks)
- d) The binary tree in **Figure 3-1** represents a dataset of integers. Using this tree, determine the traversal outputs.

**Figure 3-1: Binary tree**

- (i) Pre-order (3.5 marks)
- (ii) Post-order (3.5 marks)

[Total: 25 marks]

Question 4

A company network graph is shown in **Figure 4-1**, where each node represents a computer and each edge represents a direct connection.

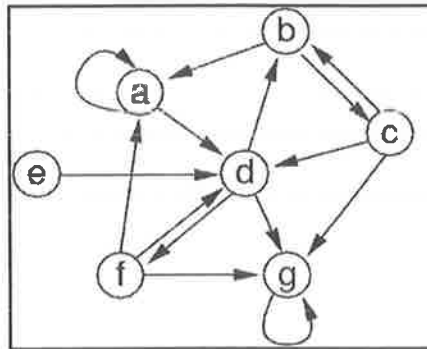


Figure 4-1: Graph of a company network

- a) Using the graph provided in **Figure 4-1**:
- Draw the adjacency matrix for the graph in **Figure 4-1**. (7 marks)
 - Write the adjacency list for the graph in **Figure 4-1**. (7 marks)
- b) A graph is a mathematical structure that represents relationships among entities. Provide **THREE (3)** applications of graphs in the real world. (6 marks)
- c) Garbage collection is an important aspect of Java memory management. Classify **TWO (2)** advantages of garbage collection and identify **ONE (1)** type of garbage collection. (5 marks)
- [Total: 25 marks]